Digital Control

Engineering Technology

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Digital Control (A11988)

Short Title:	Digital Control		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Automotive, Automation and IoT		
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Credits	5	**Level:**	Advanced

Description of Module / Aims

This course aims to introduce modern techniques used in the analysis and design of digital control systems. These techniques will be demonstrated using examples of practical control systems and will be implemented by the student using MATLAB and SIMULINK. The course should give the student the ability to bring modern control techniques to an industrial setting or perform research in state-of-the-art control methods.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (`WD_EAUTO_B`)	4 / 8 / M
`ENGR-0021`	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	4 / 8 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	4 / 8 / M
`ENGR-0021`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	4 / 8 / E
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	4 / 8 / M
`ENGR-0021`	BEng (Hons) (`WD_ETRIC_B`)	4 / 8 / M
`ENGR-0021`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	4 / 8 / E

Indicative Content

- Sampling
- Modelling of Sampled Data Systems
- Discrete System Stability Analysis
- Digital Controller Design using Transform techniques
- Direct Digital Controller Design
- Controller Design using State Space Methods
- Three Term Control: analysis and tuning
- Discrete Controller Implementation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Model discrete control systems.
2. Perform stability analysis on discrete systems.
3. Design digital controllers using transform techniques, direct and state space methods.
4. Evaluate controller designs
5. Implement Digital Controller algorithms.
6. Use MATLAB and SIMULINK as part of system analysis and controller design.

Learning and Teaching Methods

- Lectures and Practicals

- - Lectures will involvedelivery of theory and methodology with subsequent problem based learning
- - Practicals will involve use of MATLAB in analysis and design

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4
Continuous Assessment	30%	
Practical	20%	4,5,6
Lab Report	10%	4,5,6

Assessment Criteria

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- Astrom, K.J. _Computer Controlled Systems; Theory & Design_. 3rd. USA: Dover Publications, 2011.
- Franklin, G.F. _Digital Control of Dynamic Systems_. 3rd ed.. New York: Prentice Hall, 1997.
- Ogata, K. _Modern Control Engineering_. 4th ed.. New York: Prentice Hall, 2001.

Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Electronics